

ORIGINAL ARTICLE

Visual impairment and cognitive performance: A nationwide study of 1.4 million adolescents

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Abstract

Purpose: Previous research highlights the adverse effects of visual impairment (VI) on academic achievement in children, yet its impact on cognitive performance among adolescents and young adults remains under-studied. Therefore, this investigation aimed to analyse this association in a nationwide sample of Israeli adolescents.

Methods: A retrospective population-based cross-sectional study was conducted among 1,410,616 Israeli-born adolescents aged 16–19 years, who were assessed before mandatory military service between 1993 and 2017. The definition of VI was based on best-corrected visual acuity (BCVA) measurements using a standard Snellen chart. Adolescents with BCVA worse than 6/9 in either or both eyes were classified as having unilateral or bilateral VI, respectively. Cognitive performance was measured using the General Intelligence Score (GIS), based on a validated four-domain test. Relationships were analysed using regression models yielding adjusted odds ratios (ORs) for low (<−1 standard deviation [SD]) and high (≥1 SD) cognitive Z-scores.

Results: Of 1,410,616 adolescents (56.1% men), 13,773 (1.0%) had unilateral and 3980 (0.3%) had bilateral VI. Unilateral VI was associated with adjusted ORs for low and high cognitive Z-scores of 1.24 (1.19–1.30) and 0.84 (0.80–0.89), respectively. ORs were accentuated for bilateral VI, reaching 1.62 (1.50–1.75) and 0.81 (0.74–0.90) for low and high cognitive Z-scores, respectively. Cognitive performance subscores mirrored these results, with the visual–spatial functioning subtest demonstrating the greatest effect size. These associations persisted in sub-analyses restricted to adolescents with amblyopia-related VI, mild VI and unimpaired health status.

Conclusions: Visual impairment, including mild and unilateral cases, is associated with reduced cognitive performance scores assessed in late adolescence. Further research is required to gain a comprehensive understanding of the dynamics underlying this relationship.

KEYWORDS

adolescents, cognitive performance, intelligence, visual impairment

INTRODUCTION

Over the years, multiple studies have aimed to elucidate the intricate relationship between human vision and cognitive abilities.^{1–8} Research involving children has primarily concentrated on the detrimental impacts of visual impairment (VI) on academic achievement, especially during crucial phases of visual development.^{1–6} In contrast, studies within the elderly population have explored the links between deteriorating vision and cognitive decline or dementia.^{7,8} However, there is a limited body of work examining the relationship between vision and cognition among adolescents or young adults. Given that in these age groups many cognitive functions are nearing their developmental peak,⁹ and that preventable childhood ocular conditions such as amblyopia are less remediable,¹⁰ such studies are important as they may fill a gap in the literature and reveal whether the negative vision–cognition association, previously identified in early childhood, is also evident during adolescence. This is also of value considering that early-life cognitive performance is linked with long-term sequelae including increased morbidity and mortality rates^{11,12} and socio-economic success.¹³

The unique setting in Israel includes mandatory military service preceded by encompassing pre-recruitment medical and psychosocial examinations. We hypothesised that VI in adolescents is associated with decreased cognitive performance scores. Therefore, this population-based study, the largest of its kind to date, aimed to test this hypothesis and investigate the association between VI and cognitive performance scores in a sample of more than 1.4 million adolescents.

MATERIALS AND METHODS

Study population

One year before mandatory military service, Israeli adolescents around 17 years of age undergo a pre-draft screening aimed at assessing their service eligibility. This process includes the collection of sociodemographic data and medical background, a thorough medical examination conducted by a physician, as well as cognitive and behavioural assessments. Best-corrected visual acuity (BCVA) and refractive error are also measured and documented.¹⁴ The study population included Israeli-born adolescents aged 16–19 years, who were examined from 1993 to 2017. Visual acuity data were not systemically documented before 1993, and complete annual data are currently accessible up to 2017. Individuals were included in the analysis if BCVA data were available for both eyes and cognitive assessment results were recorded. Ultra-orthodox Jewish adolescents, comprising 2.9% of the target age group, were excluded due to their infrequent evaluation and distinct educational system, which has been associated with an increased prevalence of overall myopia and especially high myopia.¹⁵ In

Key points

- This population-based nationwide study involving over 1.4 million individuals revealed an association between visual impairment and decreased cognitive performance scores assessed in late adolescence.
- This association was observed in both unilateral and bilateral visual impairment cases, and persisted across scores in both verbal and non-verbal cognitive domains.
- Further research is needed to determine whether reading difficulties, cognitive deficits or a synergistic interplay of these elements underlie the observed associations.

addition, adolescents from non-Jewish minority populations were less represented in this cohort as they were not obliged to perform military service and were not systematically evaluated. The institutional review board of the Israel Defense Forces Medical Corps approved the study and participants' confidentiality was strictly maintained during data processing. Participants' informed consent was waived as the data were de-identified. The study adhered to the tenets of the Declaration of Helsinki for research involving human subjects.

Assessment of cognitive performance

Cognitive performance was assessed by the General Intelligence Score (GIS), which demonstrates a correlation >85% with the Wechsler Adult Intelligence Scale and is comparable to the Intelligence Quotient (IQ).^{11,16,17} This score has been used previously in epidemiological studies within the same cohort,¹¹ exhibiting robust external validity to other populations.¹² The Wechsler tests have been widely used in individuals with vision reduction, with relatively high levels of assessors' satisfaction rates.¹⁸ Examinees were instructed to bring their prescribed optical correction devices, if applicable, for the duration of the assessment. The GIS assessment comprises four subtests: the Otis-R, which measures the ability to understand and perform verbal instructions; Similarities-R, which assesses verbal abstraction and categorisation; Arithmetic-R, which measures mathematical reasoning, concentration and concept manipulation and Raven's Progressive Matrices-R, which measures nonverbal abstract reasoning and visual–spatial problem-solving abilities (Figure 1; additional sample questions are available in Appendix S1). The complexity of tasks within each subtest gradually increases, and examinees are required to complete each subtest within a pre-specified time limit. Examinees' scores were integrated to create a validated measure of general

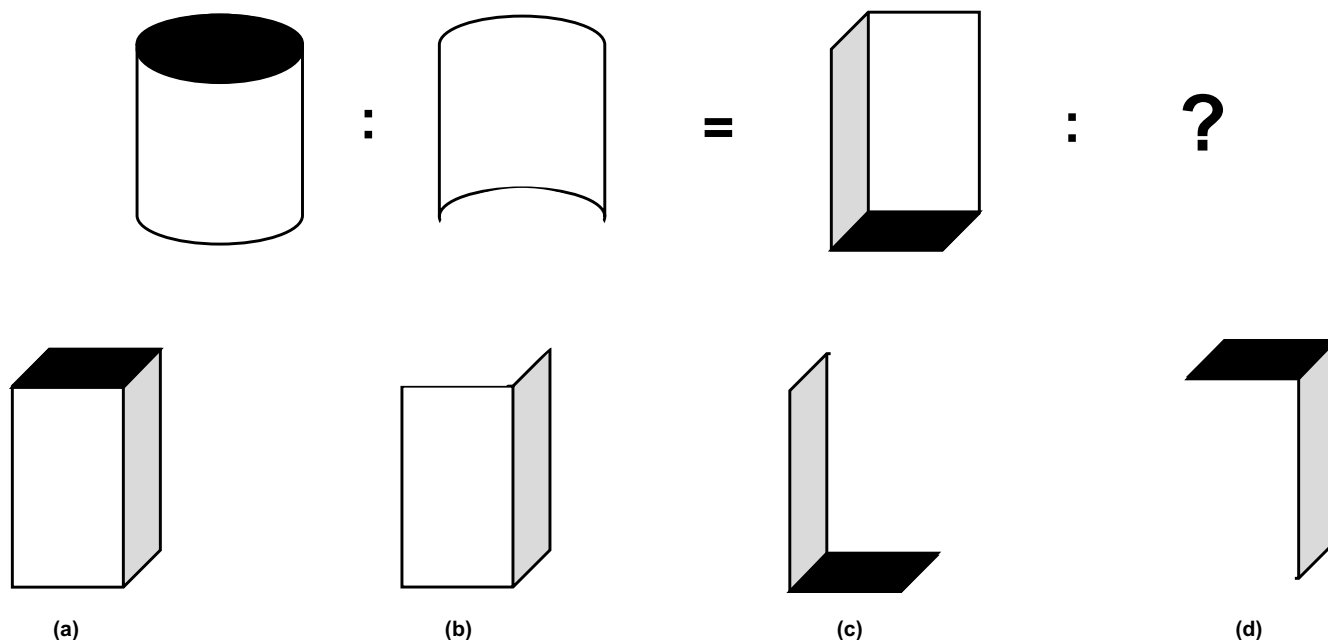


FIGURE 1 Sample question in the Raven's Progressive Matrices-R subtest. Examinees were required to assess the relationship between two given shapes, replicate the same pattern on a different shape and select the response in which the relationship between the two shapes was preserved. In this particular case, the top base of the given shape (cylinder) was removed, and the remaining shape was inverted. The correct answer was therefore (b).

intelligence (IQ), presented on a stanine scale. The GIS and each of its four components were standardised according to sex and year of performance based on all examinees who underwent cognitive assessment in a given calendar year [$Z_i = (GIS_i - GIS_{(mean)}) / GIS_{(SD)}$] (index of an individual examinee [i]), yielding annual sex-specific cognitive Z-scores. Cognitive Z-scores were categorised as low (< -1 standard deviation [SD], equivalent to an IQ < 85), medium (-1 to < 1 SD, equivalent to an IQ 85–115) and high (≥ 1 SD equivalent to an IQ > 115).

Visual acuity testing and VI definition

During pre-recruitment screening, adolescents underwent visual acuity and refraction assessments. These were carried out using a standard Snellen chart at a distance of 6 m, with each eye tested individually by a certified optometrist or ophthalmologist. Initially, unaided visual acuity was determined for each adolescent using a Snellen chart. Adolescents presenting with unaided visual acuity worse than 6/6—allowing for up to one letter error on the 6/6 line for either eye—then underwent non-cycloplegic refraction. This step was performed using an automated refractometer (models included Nikon Speedy K, Nikon.com; Topcon KR-8000, KR-7000S and earlier versions, Topcon.com). Following automated refraction, a subjective refraction was conducted for verification purposes, including measurement of the BCVA in each eye. Adolescents capable of reading all but one letter on the 6/6 line without optical correction were presumed to have a refractive error of 0 dioptres.^{14,15} Unilateral VI was

defined as BCVA worse than 6/9 in one eye only. Bilateral VI was defined as BCVA worse than 6/9 in both eyes. Adolescents with BCVA equal to or better than 6/9 in each eye were considered to have normal visual acuity. A diagnosis of amblyopia was assigned to adolescents with reduced BCVA, in the absence of other causative ocular findings, based on medical records and clinical evaluations.¹⁹ Amblyopia-related VI was classified as unilateral or bilateral using the same BCVA thresholds.

Covariates

Sociodemographic variables were collected for each individual.²⁰ Years of education were divided into complete (≥ 11 school years, the maximum possible at the time of evaluation) or incomplete (< 11 years). Socio-economic status (SES) was based on the classification method of the Central Bureau of Statistics, determined by the site of residence and categorised into three groups: low, medium and high. Country of origin was determined by the birth country of the father, or the grandfather if the father was born in Israel, and was divided into five categories: former Union of Soviet Socialist Republics (USSR), Asia (non-USSR), Africa (excluding South Africa), Western (comprised of non-USSR Europe, America, South Africa, Australia and New Zealand) and Israel. Anthropometric measurements included weight and height, measured barefoot and in light clothing by certified medics. Body mass index (BMI) was calculated as weight (kg) divided by the squared height (m^2). BMI was classified according to the age- and sex-adjusted growth charts of the United States Centers for Disease Control and

Prevention, as this categorisation was validated for the Israeli population.²¹

Statistical analysis

Categorical variables are presented as numbers and percentages, and continuous variables as means and SD. The distribution of baseline characteristics across the three cognitive Z-score groups was calculated. Multinomial logistic regression models were applied to estimate the odds ratios (ORs) and 95% confidence intervals (CIs) for low and high cognitive Z-scores across the BCVA groups. Medium cognitive Z-score and normal BCVA were set as reference categories. Models were predetermined with covariates added gradually as follows: (1) unadjusted model, (2) partially adjusted model (including sex and birth year as covariates) and (3) fully adjusted model (further adjustment for BMI, SES, level of education and country of origin). Finally, the fully adjusted model was repeated separately for each of the four cognitive assessment subscores. No statistical power calculation was performed before the study; the sample size was determined based on the available data. A two-sided $p < 0.05$ was considered statistically significant for all tests.

Subgroup and sensitivity analyses

Several pre-specified subgroup and sensitivity analyses were performed, all utilising the fully adjusted models:

1. A sex-stratified analysis to account for an effect modification of the VI-cognitive performance relationship by sex.
2. To minimise confounding by coexisting illness, which may be associated with both VI and cognitive performance, the analysis was limited to adolescents with unimpaired health status based on the pre-recruitment medical evaluation. Unimpaired health was defined as the lack of any comorbidity that required chronic medical therapy or follow-up, including central nervous system insult or neurodevelopmental disorders affecting the visual system, or any history of cancer or major surgery.²⁰
3. To minimise the potential effect of VI on task comprehension during the cognitive test administration, we focused only on adolescents with mild VI, defined as BCVA worse than 6/9 in either or both eyes, and equal to or better than 6/18 in both eyes.²² Adolescents with BCVA worse than 6/18 in either or both eyes were excluded from this analysis.
4. The analysis was limited to adolescents of medium SES to establish a more uniform socio-economic background, thereby minimising the confounding effect of low or high SES determinants on both VI and cognitive performance.

5. The analysis was limited to adolescents who had completed ≥ 11 years of education at the time of evaluation, aiming to reduce the confounding effect of low educational attainment on the association between VI and cognitive performance.
6. To account for a potential time period bias, only adolescents who were examined between 2008 and 2017 (the last 10 years of the study period) were included.
7. The analysis was limited to adolescents with amblyopia as a pre-specified cause for vision reduction. Adolescents with reduced BCVA attributed to other ocular conditions were excluded from this analysis.
8. Since higher cognitive performance has been linked with increased myopia prevalence in a similar source population,¹⁶ the analysis was limited to adolescents without myopia. Myopia was defined based on the right eye spherical equivalent (SEQ; calculated as sphere $+0.5 \times$ cylinder) ≤ -0.75 D.
9. A stricter threshold for the definition of VI was used, defined as BCVA worse than 6/12 in either or both eyes.

RESULTS

The final study sample consisted of 1,410,616 adolescents, 791,463 (56.1%) of whom were male (Figure 2). Table 1 presents the baseline characteristics of the study participants distributed according to cognitive performance score categories. The mean age of adolescents was 17.2 ± 0.4 years. Compared with the high cognitive Z-score group, adolescents with medium or low cognitive Z-scores were of lower SES (16.5%, 23.5% and 34.1% were low SES, respectively) and had a lower level of education (99.6%, 97.9% and 86.2% with complete education, respectively). Those with low cognitive Z-scores were also more likely to be underweight, overweight or obese, and of Asian or African origin. Overall, 13,773 (1.0%) and 3980 (0.3%) adolescents had unilateral and bilateral VI, respectively. Amblyopia constituted the majority of VI causes (84.2%). The proportion of adolescents with unilateral and bilateral VI increased with decreasing cognitive performance scores (Table 1).

VI and cognitive performance

Figure 3 presents the fully adjusted ORs for low and high cognitive Z-scores per BCVA group among visually impaired and amblyopic adolescents. Overall, the unadjusted and partially adjusted models were consistent with the fully adjusted model (Figure 1a). In the fully adjusted model, adolescents with unilateral VI had increased OR for low cognitive Z-scores (1.24, 95% CI 1.19–1.30) and decreased OR for high cognitive Z-scores (0.84, 95% CI 0.80–0.88). Notably, ORs for low and high cognitive Z-scores were accentuated among adolescents with bilateral VI (1.62, 95% CI 1.50–1.75 and 0.81, 95% CI 0.73–0.90, respectively).

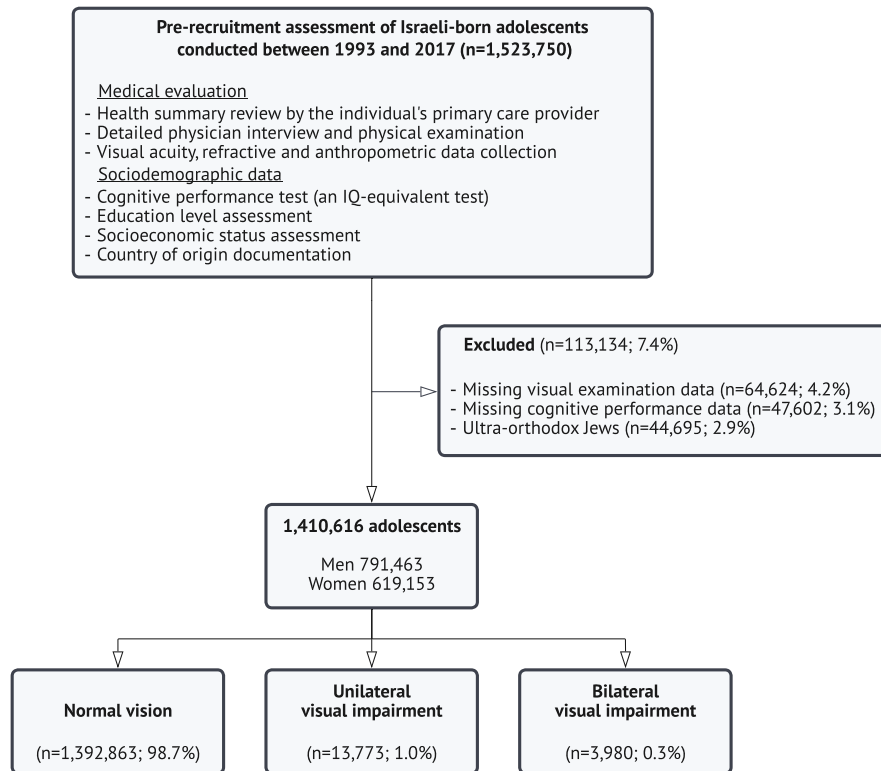


FIGURE 2 Study population build-up.

Applying the fully adjusted model for each cognitive assessment component separately revealed comparable results, with the greatest effect size noted considering the visual-spatial functioning subtest (Figure 4).

Sensitivity and subgroup analyses

The association between VI and cognitive performance persisted in various sensitivity and subgroup analyses (Tables S1 and S2). The association was evident across multiple subgroups: adolescents of both sexes, those with unimpaired health ($n=956,723$), medium SES ($n=339,431$), complete years of education ($n=1,329,368$) and without myopia ($n=1,009,528$). We also found similar estimates of association in sensitivity analyses focusing on adolescents with amblyopia-related VI ($n=1,398,122$, Figure 3), mild VI ($n=1,405,411$) and when employing a stricter BCVA threshold as a definition of VI ($n=1,410,616$).

DISCUSSION

This population-based nationwide study of over 1.4 million adolescents demonstrated that VI was associated with lower cognitive performance scores, as assessed by the GIS and its components. This relationship was evident, not only among adolescents with bilateral VI but also in those with unilateral and mild VI. ORs remained stable following gradual adjustment for multiple sociodemographic

confounders and withstood several sensitivity and subgroup analyses.

The overall reported proportion of VI in this study was 1.3% (1.0% unilateral and 0.3% bilateral); a finding aligned with previous population-based investigations worldwide.^{22–24} Growing evidence supports the paradigm that visual deficits negatively affect academic achievement in children^{1–4,6}; however, the results are mixed,^{5,25,26} and numerous methodological issues limit the ability to draw definitive conclusions from the existing literature.²⁷ For instance, the definition of VI varies between studies and may reflect either better or worse ocular function. Likewise, tools of cognitive performance assessment differ, and non-standardised, self-reported or surrogate measures of intelligence are often employed.^{2,28} Most studies have used relatively small sample sizes, ranging from dozens^{1–3} to thousands^{5,6,25} of participants, and population-based data, particularly among youth, are lacking. The current findings extend current knowledge relying on a vast population-based screening process and indicate that lower cognitive performance among visually impaired individuals may also become manifest during late adolescence.

Several investigations have addressed directly the impact of naturally occurring or simulated reduced vision on cognitive test performance, mostly in the adult population. While some exemplified null effects,^{29,30} others have demonstrated that specific cognitive domains, particularly non-verbal skills, are more susceptible to low vision.³¹ Contrary to these mixed findings, the present study showed that VI was associated with reduced cognitive

TABLE 1 Baseline characteristics of the study population.

	Low cognitive Z-score (<-1 SD)	Medium cognitive Z-score (-1 to <1 SD)	High cognitive Z-score (≥1 SD)
Participants, <i>n</i> (%)	251,918 (17.9)	915,021 (64.9)	243,677 (17.3)
Sex, %			
Men	62.0	56.9	47.1
Women	38.0	43.1	52.9
Mean age (SD) (years)	17.4 (0.5)	17.2 (0.4)	17.1 (0.3)
Mean BMI (SD) (kg/m ²)	22.5 (4.4)	22.1 (3.8)	21.8 (3.4)
BCVA group, %			
Normal	98.4	98.8	99
Unilateral visual impairment	1.2	1.0	0.8
Bilateral visual impairment	0.4	0.3	0.2
Amblyopia, %			
Unilateral	1.1	0.8	0.7
Bilateral	0.4	0.2	0.2
Residential socio-economic status, %			
High	12.9	22.9	34.2
Medium	53.0	53.6	49.3
Low	34.1	23.5	16.5
Years of education, %			
Complete (≥11 years)	86.2	97.9	99.6
BMI category, %			
Underweight	7.8	5.7	4.9
Normal weight	71.4	77.3	81.7
Overweight	12.0	10.9	9.4
Obese	8.8	6.1	4.0
Country of origin, %			
Israel	19.6	21.6	21.8
USSR	7.5	8.3	9.4
Asia	24.8	22.6	16.1
Africa	37.4	23.5	12.9
West countries	10.7	24.0	39.8

Abbreviations: BCVA, best-corrected visual acuity; BMI, body mass index; SD, standard deviation; USSR, former Union of Soviet Socialist Republics.

performance scores across all four examined cognitive domains, with the visual-spatial subtest displaying the strongest correlation with VI. These results are supported by evidence suggesting a shared visual form processing mechanism linking reading abilities and arithmetic skills,³² as well as a connection between visual information processing and mathematical competence.^{1,3,33} Visual-spatial skills and arithmetic performance also interact with each other, as suggested by commensurate regions of brain activation in functional magnetic resonance imaging (fMRI)³⁴ and mutual genetic factors.³⁵

Alternatively, visually impaired adolescents might attain lower cognitive performance scores because of reading difficulties, thereby possibly failing to demonstrate their full cognitive potential in a time-limited test. Indeed, children with visual deficits demonstrate lower literacy

scores and decreased reading proficiency relative to their peers.^{2,28} Amblyopic children also present with slow reading, which has been attributed to the 'crowding' phenomenon, poor eye fixation and an increased number of saccades.³⁶ Recent research indicates that the presence of amblyopia itself, rather than the extent of visual deficit, is correlated with decreased reading speed, both under binocular and fellow-eye reading conditions.¹⁰ The current data potentially corroborate this pattern, as a consistent relationship was observed between VI and reduced cognitive performance scores even among adolescents with only mild VI (BCVA equal to or better than 6/18), as well as in cases where visual acuity lay within the normal range in one eye (unilateral VI).

It should be acknowledged that all four cognitive subtests employed in this study entailed a certain level of

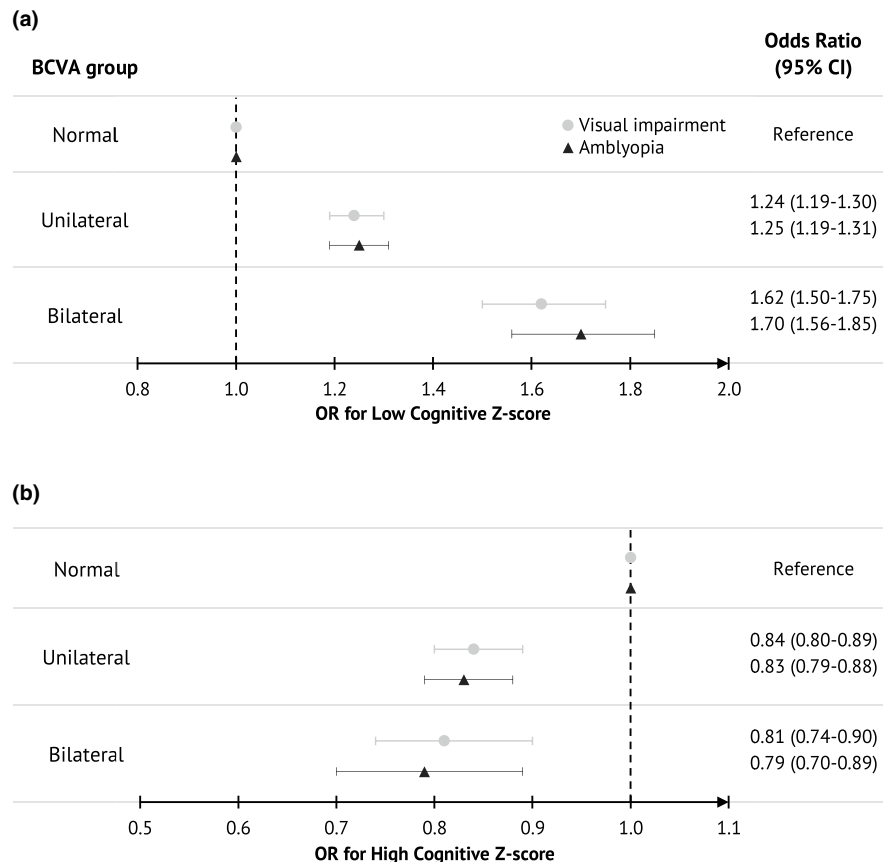


FIGURE 3 The association of visual impairment and amblyopia with (a) low and (b) high cognitive performance in fully adjusted models (Adjusted for sex, birth year, body mass index, residential socio-economic status, level of education and country of origin). BCVA, best-corrected visual acuity. CI, confidence interval. OR, odds ratio.

reading (Appendix S1), which, in turn, limited our capacity to attribute the observed decrease in cognitive performance to factors independent of reading difficulties. This complexity was further echoed in a meta-analysis emphasising the heterogeneity of cognitive profiles in individuals with reading difficulties, as well as the bidirectional pathways linking cognitive abilities and reading development.³⁷ Therefore, the observed decrease in cognitive performance scores could be argued to represent a broader outcome of intertwined influences of reading skills and cognitive abilities, within the context of VI. This hypothesis, however, requires confirmation through dedicated studies within the fields of educational psychology and neuropsychological assessment.

While we could not confirm the exact underlying mechanism of the observed link between visual acuity and cognitive performance, this study may hold significance for public health. The findings suggest the potential benefit of incorporating vision-adjusted time extension as a compensatory intervention to address the challenge of slow reading among visually impaired youth. Importantly, the findings also prompt the re-evaluation of formal vision thresholds regarding eligibility for academic accommodations,¹⁰ especially given the decrease in cognitive performance scores seen even among adolescents with mild VI.

Regarding modifications to standardised cognitive assessments, various adapted versions utilise haptic modalities intended to address the needs of visually impaired children and adolescents adequately.³⁸ That being said, any testing adaptations must first be validated to allow comparability with other standardised instruments.

This study has several limitations. First, the cross-sectional study design prohibits inferring a causal relationship between VI and cognitive performance. Second, we lacked data on the allocation of partial credit for visual acuity measurements. However, this limitation is expected to introduce a non-differential misclassification bias. Third, the use of the Snellen chart might have led to inaccurate assessments of low vision levels. Our definition of VI, that is, identifying visual acuity worse than 6/9, likely mitigates this limitation. Fourth, the assumption that 6/6 vision precludes uncorrected refractive error may overlook the impact of significant hyperopia on sustained near-vision tasks, potentially leading to an underestimation of the observed associations in this subgroup. Fifth, our data set lacked temporal granularity to capture the exact timing of VI onset and the progression of vision deterioration. We also lacked data on potential confounders such as parental cognitive assessment and former academic achievement. Nevertheless, the results withstood

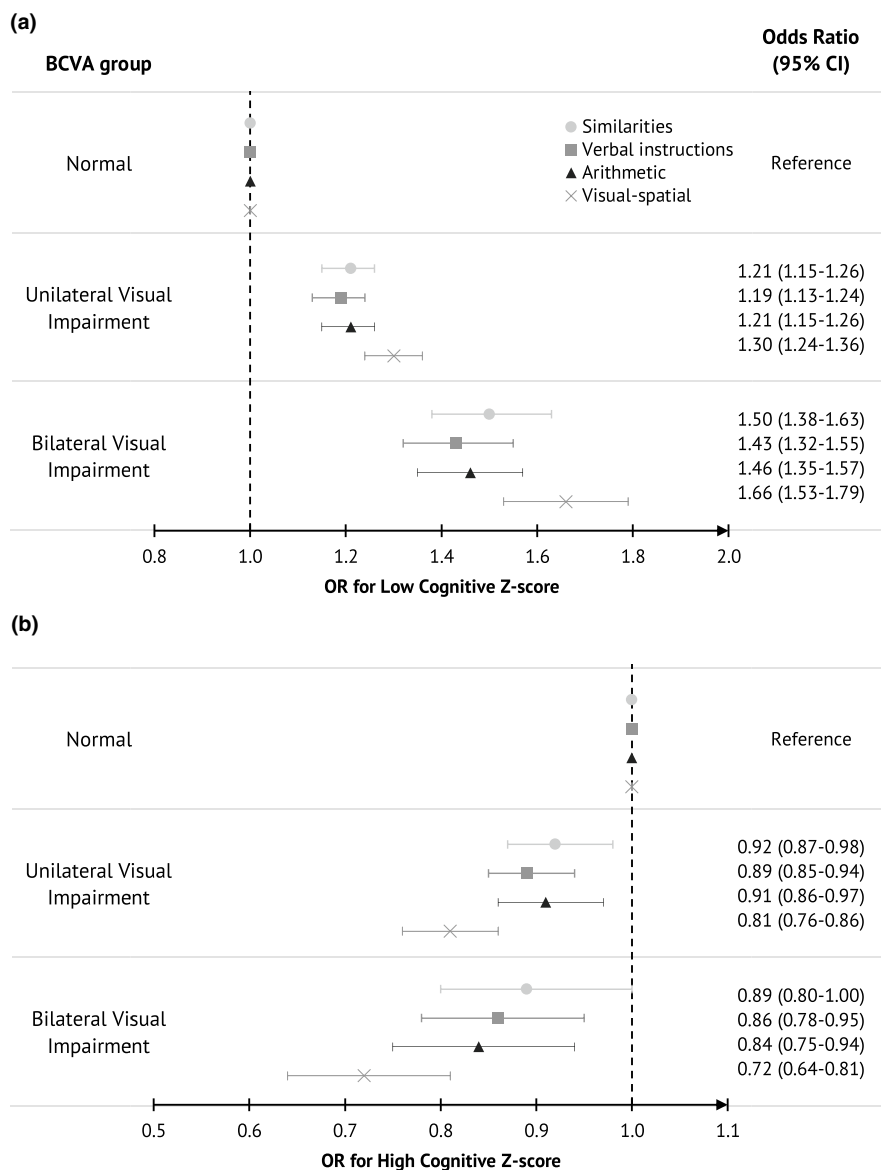


FIGURE 4 The association of visual impairment with (a) low and (b) high cognitive performance by different subtests in fully adjusted models (Adjusted for sex, birth year, body mass index, residential socio-economic status, level of education and country of origin). BCVA, best-corrected visual acuity. CI, confidence interval. OR, odds ratio.

several sensitivity and subgroup analyses, including those restricted to adolescents with unimpaired health status, amblyopia-related VI and without myopia, collectively displaying parallel results. Sixth, by focusing on visual acuity measurements, we might have overlooked other vision assessment aspects like visual fields, contrast sensitivity and binocularity. While these factors may contribute to the heterogeneity of visually impaired adolescents, visual acuity is commonly used as a preferred measure of vision in large epidemiological studies.²²⁻²⁴ Finally, as noted, we could not ascertain whether decreased cognitive performance scores observed in visually impaired adolescents indicated genuine cognitive deficits or rather an effect related to reading difficulties. As research indicates a multi-dimensional relationship between cognitive development

and reading abilities,³⁷ these findings call for further longitudinal studies to better understand these interrelations among youth with VI. The strengths of this investigation include the reliance on a large national registry encompassing three decades, the systematic and screening-based collection of sociodemographic and medical data including BCVA measurements and the battery of cognitive tests yielding the GIS.

In conclusion, this study highlights an association between VI and reduced cognitive performance scores in a nationwide sample of adolescents, as measured by the GIS assessment. Importantly, this association was consistently evident in cases of both unilateral and mild VI. Further research is essential for determining the exact causal pathways contributing to the observed relationship

and guiding tailored adaptations in cognitive assessment methodologies for youth with VI.

AUTHOR CONTRIBUTIONS

Itay Nitzan: Conceptualization (lead); data curation (equal); formal analysis (equal); investigation (equal); methodology (lead); visualization (lead); writing – original draft (lead); writing – review and editing (equal). **Estela Derazne:** Data curation (equal); formal analysis (equal); writing – review and editing (supporting). **Arnon Afek:** Project administration (equal); supervision (equal); writing – review and editing (supporting). **Adi Einan-Lifshitz:** Supervision (equal). **Yair Morad:** Supervision (equal). **Claudia Yahalom:** Investigation (supporting); methodology (supporting); writing – review and editing (supporting). **Alon Peled:** Investigation (equal); methodology (equal); project administration (equal); supervision (equal); writing – review and editing (equal).

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None.

CONFLICT OF INTEREST STATEMENT

The authors report there are no competing interests to declare.

DATA AVAILABILITY STATEMENT

The current data set is subject to military restrictions, and therefore its availability is limited. Data requests or queries may be addressed to the corresponding author.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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